This relationship may be presented in the following general form

$$\begin{array}{ccccccc} y_1 & y_2 & y_3 & y_4 & y_5 & \cdots\cdots & y_n \cdots\cdots \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & & \uparrow \\ 1 & 2 & 3 & 4 & 5 & \cdots\cdots & n \cdots\cdots \end{array}$$

The number y_n is the nth term of the sequence, and the whole sequence is sometimes denoted by a symbol (y_n).

Reader: We have been given a somewhat different definition of a sequence: a sequence is a function defined on a set of natural numbers (integers).

Author: Well, actually the two definitions are equivalent. However, I am not inclined to use the term "function" too early. We will deal with them later. Anyway, the above definition of a sequence is quite correct.

• Getting back to our examples of sequences, let us look in each case for an analytical expression (formula) for the nth term. Go ahead.

Son: Seq 1 $\to 2^{n-1}$ Seq 2 $\to 2n+3$ Seq 3 $\to n^2$ Seq 4 $\to \sqrt{n}$

Seq 5 $\to \frac{n}{n+1}$ Seq 6 $\to 2-2(n-1)=4-2n$ Seq 7 $\to \frac{1}{n}$

Seq 8 $\to$ If n is even, or $n=2K$, $K\in N$, then $y_n=\frac{1}{n}$

If n is odd, or $n=2K-1$, $K\in N$, then $y_n=n$

Seq 9 $\to$ If n is even, or $n=2K$, $K\in N$, then $y_n=-\frac{1}{(n-1)}$

If n is odd, or $n=2K-1$, $K\in N$, then $y_n=\frac{1}{n}$

Reader: Oh, this is not difficult. In example (1), it is $y_n=2^{n-1}$. In (2), it is $y_n=2n+3$. In (3), it is $y_n=n^2$. In (4), it is $y_n=\sqrt{n}$. In (5) it is $y_n=1-\frac{1}{n+1}=\frac{n}{n+1}$. In (6) it is $y_n=4-2n$. In (7), it is $y_n=\frac{1}{n}$. In the remaining three examples I just do not know.

Author: Let us look at example (8). One can easily see that if n is an even integer, then $y_n=\frac{1}{n}$, but if n is odd, then $y_n=n$. It means that

$$y_n=\begin{cases}\frac{1}{n} & \text{if } n=2K \\ n & \text{if } n=2K-1\end{cases}$$

Reader: Can I, in this particular case, find a single analytical expression for y_n?